



Influences of tip clearance on flow and heat transfer characteristics of open type micro pin fin heat sink

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ABSTRACT

Microchannel heat sink without any tip clearance i.e. completely closed microchannels have been focused and widely studied. In the present work, detailed comparative analysis has been performed between open and closed micro pin fin configurations of the heat sink. Keeping the constant channel height of 2.0 mm, three different micro pin fin heat sinks with tip clearance of 0%, 25% and 50% of channel height were fabricated and analyzed. Square cross-section fins are considered and arranged in inline fashion. An additional plain channel (without any pin fin) is also examined for the sake of comparison and completeness of the study. The experiments have been conducted using deionized water as the working fluid and copper as microchannel substrate. Present work aims to recognize the importance of tip clearance and quantify its suitable value that facilitates better heat transfer rate. Numerical predictions further elaborate the physical insight of the coolant flow behavior evolved due to the presence of tip clearance in the heat sink. Findings of the present study indicate that tip clearance of 50% results in lesser heat transfer coefficients due to considerably lower net convective surface area. Whereas, tip clearance of 25% has shown superior thermal performance amongst all the configurations explicitly at Reynolds number more than 350. In addition to the net convective surface area, major factors that promote enhanced heat transfer are dispersed fluid flow behavior in the confined space of tip clearance which propagates thinner boundary layer, three dimensionalities in flow and augmented fluid mixing. Moreover, heat sink with tip clearance has also shown lesser pressure drop compared to closed microchannel due to reduced flow obstructions. It has been concluded that tip clearance may have positive impact on overall thermal performance but up to the certain limit.

1. Introduction

Compact heat sinks made-up of mini or microchannels have shown their heat transfer potential in diverse fields of engineering and technology where high heat flux dissipation is required. Subsequently, mini or microchannel heat sinks have secured its applications in various fields such as electronic, chemical processes, microfluidics devices, hybrid vehicle, small scale power plants, refrigeration and air conditioning (RAC), Micro-electro mechanical systems (MEMS), high process computing, and many more. Among the popular cooling techniques i.e. heat pipe, jet impingement, thermoelectric coolers, electro-osmotic pumping, scope and applicability of microchannel heat sinks have been mostly deliberated in highly compact electronic devices and high performance servers [1–4]. Consistent trend of miniaturization along with high performance requirements sought from an electronic device

has drastically increased the heat generation per unit volume of the components. Moreover, it has already been projected that shortly, in coming future, 1000 W/cm² heat may generate in compact electronic devices [5]. Hence, challenges are persistent to dissipate high density heat so that safe, efficient and long-lasting usability of such devices could be assured. Traditional or even current cooling methods like, natural or forced air cooling may not be sufficient to meet the future needs henceforth, leading industries and manufacturers have started pondering to equip high heat generating devices with emerging liquid cooling technologies [6]. Microchannel heat sinks with single-phase liquid coolant are one the most prominent candidates amongst such emerging cooling solutions and certainly secured its commercial application mostly as an external cooler for electronic devices [2]. However, this technology is not enough matured and needs consistent efforts to fulfil the upcoming cooling requirements. Two important aspects where

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Table 1

Selected works pertaining to heat sink with TC for single phase coolant flow.

Author	Year	Type of work	Fin type	Parameters considered	Key observation
Rozati et al. [40]	2008	N	Circular	$TC/D_f = 0-0.4$	Reported $TC/D_f = 0.3$ as most efficient for low Re and while $TC/D_f = 0.2$ and 0.3 have shown similar performance at higher Re
Mei et al. [39]	2014	E & N	Circular	$H_{ch} = 1, 1.2, 1.5, 2$ mm, $H_f = 1$ mm	Proposed correlations for heat transfer and friction factor accounting the TC effect
Far et al. [41]	2015	N	Oblique	$TC/W_c = 0-0.74$	Provides optimum range of TC/W_c for water and nano encapsulated phase change material slurry
Prajapati [32]	2019	N	Rectangular	$H_{ch} = 1$ mm $H_f = 0.4-1.0$ mm	The TC of 15–20% of channel height has better thermal performance
Kadam et al. [33]	2019	E	Circular	$H_{ch} = 700$ μ m $H_f = 310$ μ m	Open microchannel heat sink has comparable heat transfer characteristics with the closed microchannel with less pressure drop
Bhandari and Prajapati [9]	2021	N	Square	$H_{ch} = 2$ mm $H_f = 0.5-2.0$ mm	Fin height of 1.5 mm has shown superior performance than all considered cases
Tabkhi et al. [34]	2020	E & N	Circular	$H_{ch} = 200$ μ m $H_f = 100, 155$ and 170 μ m	Augmentation in heat transfer was shown at higher TC
Bhandari and Prajapati [35–38]	2021	N	Square	$H_{ch} = 2$ mm $H_f = 0.5-1.5$ mm (stepped manner)	Stepped fin arrangement along with decreasing TC shows better thermal performance than uniform pin fin height

E: experimental, N: numerical.

researchers are putting their sincere attention are (i) packaging and fitting of liquid cooling heat sink in the electronic devices and (ii) thermal performance enhancement of the heat sink with some innovative design and techniques. Since the commencement of the novel idea by Tuckerman and Pease [7], literature has witnessed the enormous works pertaining to improve the thermo-hydraulic performance of the microchannel heat sinks.

In order to enhance the thermal performance of the microchannel heat sinks, the most prevalent amendments apparent from the literature are geometrical modification, variation in working fluids, coolant flow behavior (single-phase or flow boiling), operating conditions and surface texture of the heat sink [8–11]. However, rectangular parallel channel is widely recognized and accepted configuration among the researchers of this field [12]. It is attributed to its simple geometry, easy to manufacture, uncomplicated coolant flow behavior and comparatively lesser pressure drop. Likewise, pin fin microchannel heat sink is also gaining attention in the research fraternity. It has been identified that pin fin profile facilitates increased convective surface area, better fluid mixing, thinner and prolonged developing boundary layer consequently; it is capable of dissipating higher heat [13]. To further enhance the performance of micro pin fin heat sinks, various studies have been performed where main interests are to improve pin fin shape, dimensions, pin fin density or porosity etc. [6].

It has been noticed that pin fin shape or channel geometry certainly affects the overall performance of the heat sink. Some of the commonly investigated pin fin shapes are circular, rectangular, square, cone, diamond, triangular, pentagonal, hexagonal, trapezoidal [13–17]. Izci et al. [16] numerically studied six different pin fin profiles under same chord thickness/diameter constraint. Rectangular pin fin shows maximum Nusselt number (Nu) and pressure drop but cone shaped pin fin has highest thermal performance index. In a similar type of experimental and numerical work, Yang et al. [13] found hexagonal pin fin performing best among five different types of the pin fin heat sinks. Hua et al. [18] conducted experimental study on micro pin fin heat sink with unlike pin fin shapes. They concluded that difference in heat transfer rate among the fin profiles is noticeable at Reynolds number (Re) greater than 100. Moreover, elliptical pin fin was reported with highest heat transfer performance under constant coolant flow rate. They also proposed correlations between Nu and Re for the considered pin fin shapes. Based on experimental work on circular and square pin fin heat sinks, Prasher et al. [19] also proposed correlations for average Nu and friction factor. In addition to above discussed pin fin shapes, some unusual pin fin designs were also proposed in microchannel heat sink like piranha, open ring etc. [20,21]. Furthermore, it has been observed that arrangement or distribution of the fins in the heat sink affects the

coolant flow behavior and hence heat transfer rate. Inline and staggered distribution of the fins are two common considerations mostly opted in the studies. Staggered arrangement favors slightly higher heat transfer than inline arrangement for a given mass flow rate [22] but at the expense of increased pressure drop. While working on oblique and staggered arrangement of pin fin, Shafeie et al. [23] found nominal difference in heat transfer rate due to changing distribution pattern of fins.

Pin fin density or microchannel porosity is another factor explored in the literature which basically depends on number of fins present in the heat sink. Pin fin density is defined as the total volume of the fins present in the microchannel. Whereas, porosity is the volume through which coolant flows in the entire structure. Hua et al. [18] reported that higher fin density has shown lesser flow performance. Hernandez et al. [24] proposed variable fin density microchannel along the flow direction for heat transfer enhancement and mitigation of surface temperature non uniformities. They concluded that the proposed configuration was able to maintain the surface temperature in the allowable range for IC chips with reduced pressure drops. Similar kind of conclusion was also reported by Villarubi et al. [25] in their experimental and numerical works. Zhao et al. [26] observed that rotation angle of 30° and porosity of 0.75 attains better thermal performance. Rasouli et al. [27] claimed that higher aspect ratio and pitch ratio resulted in higher Nu in the considered range of Re . Similarly Xie et al. [28] studied the influences of sidewall proximity and pointed out that side gap to pin fin diameter (D_f) ratio of 0.9, 1.0 and 1.1 have comparatively enhanced overall thermal performance.

It is worth mentioning that studies on microchannel are mostly limited to closed microchannel heat sinks having fin height (H_f) and channel height (H_{ch}) are always equal. In these configurations, fin top surface remains in contact with the top cover wall therefore, usually adiabatic boundary condition is considered. While, tip clearance (TC) evolves when fin height is kept slightly lesser than channel height. Hence, top surface of the fins encounters convective boundary condition. Recently, studies on TC have been emphasized in the literature and the researchers have coined a new term namely open microchannel if a TC exists in the heat sink. Originally, this design was emerged to overcome the issues of two-phase flows in microchannels. Cooke and Kandlikar [29] used this configuration for enhancing pool boiling performance while Kandlikar et al. [30] employed it in the flow boiling experiment. Yin et al. [31] performed visualization study on open microchannel and observed that the additional space above the microchannel provides less resistance for bubble expansion and causes two phases stratification. They also concluded that open configuration of microchannel helps in suppressing the flow boiling instabilities. In order

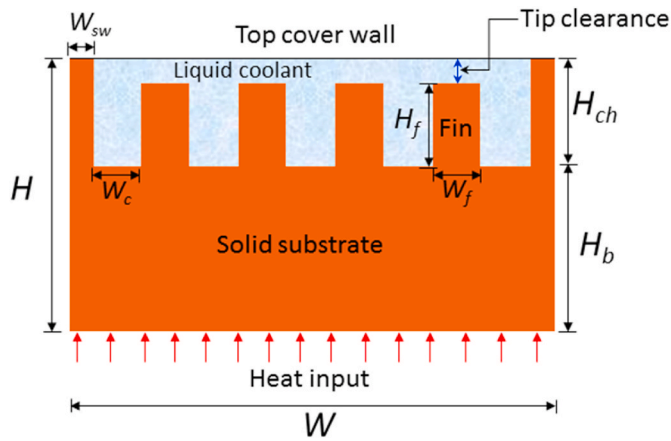


Fig. 1. Schematic of the heat sink with TC and other dimensional parameters.

to identify the potential of open microchannel heat sink with single phase coolant flow, few studies have been reported in the literature. Prajapati [32] has numerically analyzed the rectangular parallel microchannel heat sinks with variable fin height and observed that heat sink with fin height $\approx 80\text{--}85\%$ of channel height has optimum performance compared to other configurations. Kadam et al. [33] also worked on plain and extended open microchannel. Bhandari and Prajapati [9] conducted a numerical study on open microchannel with varying fin height and concluded that 15–25% of TC supports better heat transfer rate compared to the heat sink with no TC. Tabkhi et al. [34] conducted numerical and experimental study on microchannel heat sink with variable TC and pointed out that existing TC enhances the heat transfer due to increased mixing and three dimensionality nature of the coolant flow. Bhandari and Prajapati [35–38] numerically studied the effect of array of stepped micro pin fin with varying TC. Authors observed that array of decreasing TC shows enhanced thermal performance than its reverse pattern. Table 1 [9,32–41] highlights some of the studies performed with TC using single phase fluid flow in the micro pin fin heat sink. It has been found that microchannel heat sink with optimum TC has the potential to dissipate higher heat compared to heat sink without tip clearance. Moreover, flow resistance offered by such configurations is also less. Nevertheless, very limited works have been reported on such configurations of the heat sink and to the author's best knowledge, inclusive experimental study is not available in the literature. Prevailing gap in the literature has motivated us to carry out comprehensive

Table 2
Detailed dimensions of the pin fin heat sink.

Parameter	Value (mm)
Total length of the heat sink (L)	27.0
Total width of the heat sink (W)	10.0
Heat sink inlet width (W_{ch})	9.0
Pin fin height (H_f)	1.0, 1.5 and 2.0
Side wall thickness (W_{sw})	0.5
Total height of the heat sink ($H = H_{ch} + H_b$)	6.0
Heat sinks base thickness (H_b)	4.0
Fin base dimension ($L_f \times W_f$)	1×1
Individual channel width (W_c)	1.0
Gap between two successive fins in all directions	1.0
Total number of fins rows	4.0
Total number of fins in each row	12.0

experimental and numerical work on heat sink with variable pin fin TC.

In our previous numerical work [9], an effort was made to recognize the most appropriate heat sink configuration with optimum TC that exhibits superior thermo-hydraulic performance. In order to substantiate the numerical findings of the previous work [9] and to further establish the heat transfer potential of the heat sink with suitable TC, present study has been undertaken to fulfil the following objectives. (i) To evaluate the thermo-hydraulic performance of open type microchannel heat sinks with TC of 50% and 25% and (ii) compare the results obtained from both configurations with completely closed heat sink ($TC = 0$) and plain channel. Moreover, this study also includes numerical investigation that aims to thoroughly understand the coolant flow behavior and inside flow physics evolved due to the presence of TC.

2. Experimental setup and procedure

2.1. Fabrication of micro pin fin heat sink

Three different configurations of the square pin fin heat sink i.e. (i) $H_f/H_{ch} = 1.0$ ($TC = 0$) (ii) $H_f/H_{ch} = 0.75$ ($TC = 25\%$) (iii) $H_f/H_{ch} = 0.5$ ($TC = 50\%$) and a plain channel (without any fin) have been fabricated on four separate copper blocks. Each copper block has equal footprint area of length $\times$ width = $27 \times 10 \text{ mm}^2$. All these heat sinks are created using computer numerical control milling machine with vertical machining center. In order to provide required TC, initially, facing operation was done on copper block followed by milling operation to finally generate the channel and pin fin profiles. Complete nomenclature and representation of the various dimensional parameters have been

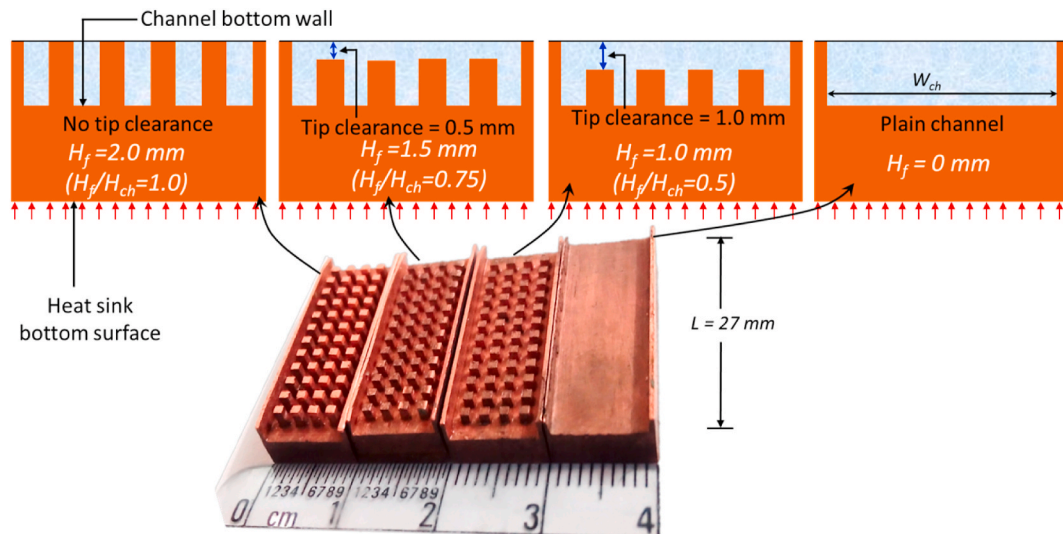


Fig. 2. Photograph of all four heat sinks with appropriate cross-sectional sketches.

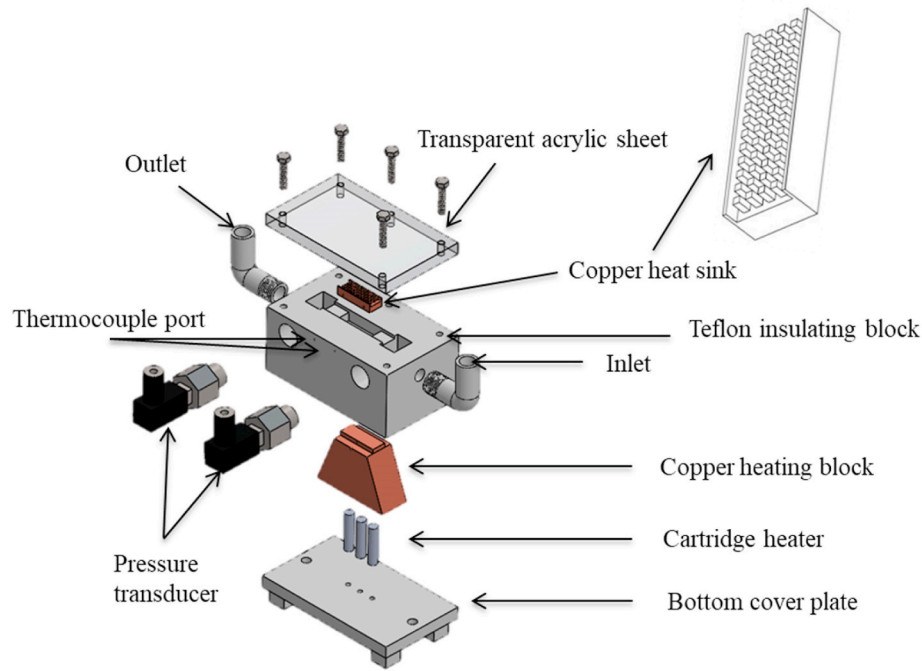


Fig. 3. Exploded view of the test assembly.

shown in Fig. 1. TC facilitates additional space between fin's top surface and cover wall of the heat sink. Unlike conventional heat sinks having $TC = 0$, top surfaces of the fins also participate in the heat transfer process if TC is provided.

In addition to three finned heat sink profiles, an additional plain channel has also been fabricated for the sake of comparison and completeness of the study. Fig. 2 clearly shows the photograph of all four channel profiles along with their suitable sketches. It should be noted that channel height in all the configurations is equal i.e. $H_{ch} = 2.0$ mm however, H_f/H_{ch} is ranging from 0.5 to 1.0 due to varying fin height. Detailed dimensions of the micro pin fin heat sinks are tabulated in Table 2. In order to embed the thermocouples to measure wall temperature, two holes of 1 mm diameter and 3 mm depth were drilled at the side face of each configuration. Both holes are located 2 mm below the bottom wall of channel however; relative positions of the holes are 5

mm and 22 mm from the inlet.

2.2. Assembling of the test section

Test section assembly consists of six major components that are copper heating block, cartridge heater (make Marathon heater), Teflon housing block, pressure transducers (make Elcom) thermocouples (make Tempens Instruments) and pin fin microchannels. Two cover plates (top and bottom) made of acrylic sheet are used to support and fix the complete assembly with nut and bolts. Top cover plate also facilitates in flow visualization of the coolant and helps to identify any leakage, trapped air and unstable flow conditions. Exploded view of the complete test section is shown in Fig. 3. Teflon housing block is the core component that holds all other required attachments. It is having strategically located inlet and outlet plenums along with shallow passage to ensure

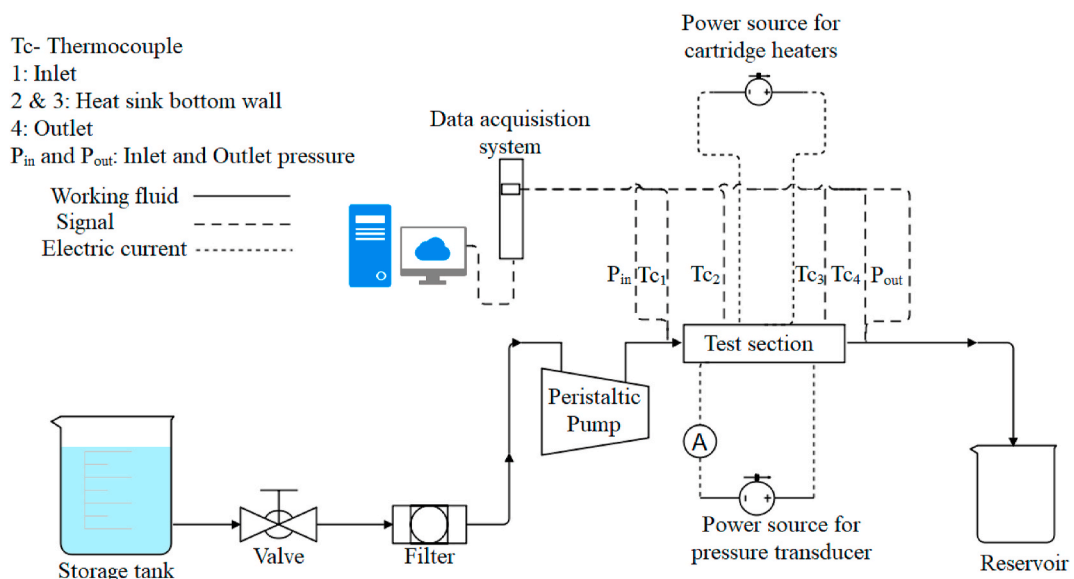


Fig. 4. Schematic of experimental setup.

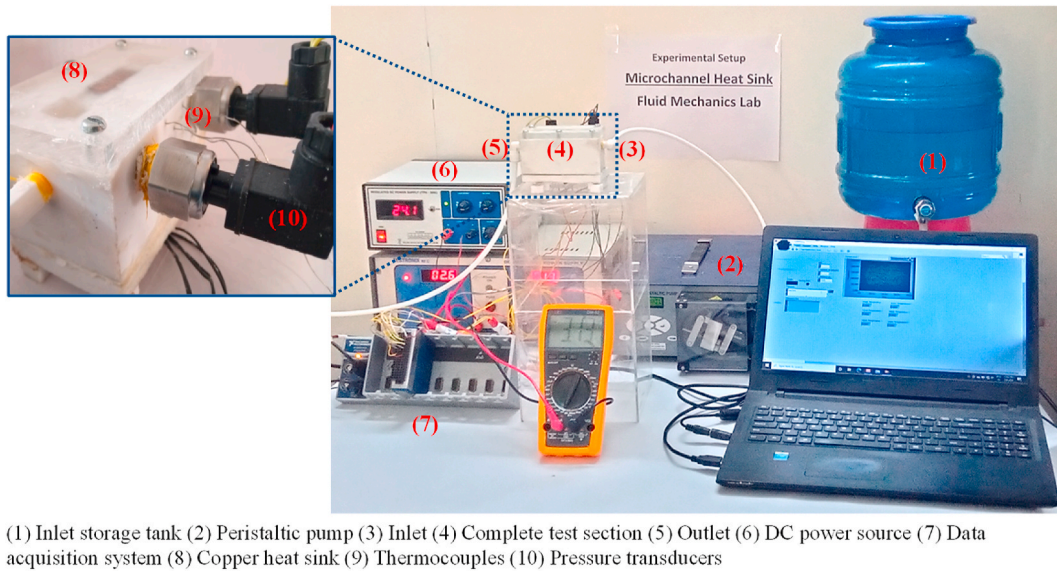


Fig. 5. Photograph of the experimental setup of pin fin microchannel.

smooth flow of the coolant through the channel. Moreover, housing block also accommodates two pressure sensors connected respectively in the inlet and outlet plenums, thermocouples and coolant inlet and outlet passages. Heating block made of copper is also inserted in it from the bottom. It should be noted that heating block is created using wire cut Electrical discharge machining; it houses three high heat density cartridge heaters and provides uniform heat to the pin fin heat sink. It is tapered in geometry with bottom area $54 \times 20 \text{ mm}^2$ and top surface area of $27 \times 10 \text{ mm}^2$. Tapered shape was provided to allow it for better and concentrated heat conduction towards the top surface. As shown in Fig. 3, top surface of the heating block and bottom surface of the heat sink come in direct contact with each other. In order to avoid any air gap, thermal paste is applied at the interface of both the surfaces to ensure proper heat conduction to the heat sink. High watt density cartridge heater (each 250 W rating) is inserted in three different holes drilled at the bottom of heating block. All three holes are equidistant from each other so that uniform heat flux can be generated in the heat source. It can be seen that; heating block is placed inside Teflon housing block. To reduce the heat loss, an insulator namely glass wool was filled in the gap available between heating block and housing assembly.

Coolant initially enters into the inlet plenum followed by shallow passages and finally to the microchannel. Pressure transducers generate output signal in the form of current (4–20 mA), which is further calibrated to corresponding pressure data. To avoid any possibility of leakage of coolant or air entrainment in the test section, a very thin layer of transparent silica paste was applied in between top acrylic cover and Teflon block.

2.3. Experimental setup and procedure

A schematic diagram of the experimental setup is shown in Fig. 4. Deionized water is stored in the storage tank which is maintained at atmospheric pressure. A peristaltic pump circulates and precisely measures the coolant flow rate passing through the test section. While passing through the test section, coolant takes heat from the heated microchannel and discharges it to a reservoir. Temperature and pressure data is recorded using National Instruments data acquisition system and Lab view software. Schematic loop also depicts the signals and electric current supply lines. Direct current (DC) power supply is essentially required for the cartridge heater and pressure transducer to actuate it. It is worth mentioning that during the experimentation, two parameters i. e. heat flux (q) and coolant flow rate have been varied. Coolant inlet

Table 3

Operating ranges in the experimental work.

G (kg/m ² s)	q (kW/m ²)	Re	Inlet temperature (T_m)
23–138	30–60	90–570	30 °C

temperature remains fixed to 30 °C with variation of ± 1 °C. Before commencement of the experiments, deionized water was boiled vigorously to remove dissolved air if any. Subsequently, it allows to cool down to inlet temperature of 30 °C.

In the beginning, the pump was switched on and coolant was allowed to flow for 3–4 min to remove any trapped air in the flow loop and the microchannel. After the stabilization of the coolant flow at the desired rate, controlled DC power was supplied to the cartridge heater and also to the pressure transducer (24 V) as actuation energy. Simultaneously, data acquisition system was also put on to record temperature and pressure data. For each test run, steady state was achieved in 30–35 min. For the same flow rate, heat input to the test section was further increased with variac and data was continuously recorded. Fig. 5 clearly shows the photograph of the complete experimental setup with enlarged view of the test section. In the similar way, all experimental runs were conducted. Operating ranges of different parameters are shown below in Table 3.

2.4. Data reduction

During experiments, cartridge heater was given variable voltage (V) and current (I) from the DC power source. Hence, total heat input (Q) supplied to the test section is calculated using Eqn. (1).

$$Q_{total} = V \times I \quad (1)$$

q supplied to the heat sink was calculated on the basis of bottom surface area or footprint area ($A_{bs} = L \times W = 27 \times 10 \text{ mm}^2$) and is expressed as Eqn. (2).

$$q = \frac{Q_{total}}{A_{bs}} \quad (2)$$

It should be noted that total heat generated from the cartridge heaters is not fully transferred to the working fluid due to associated heat losses. The effective heat being supplied to the working fluid is given by:

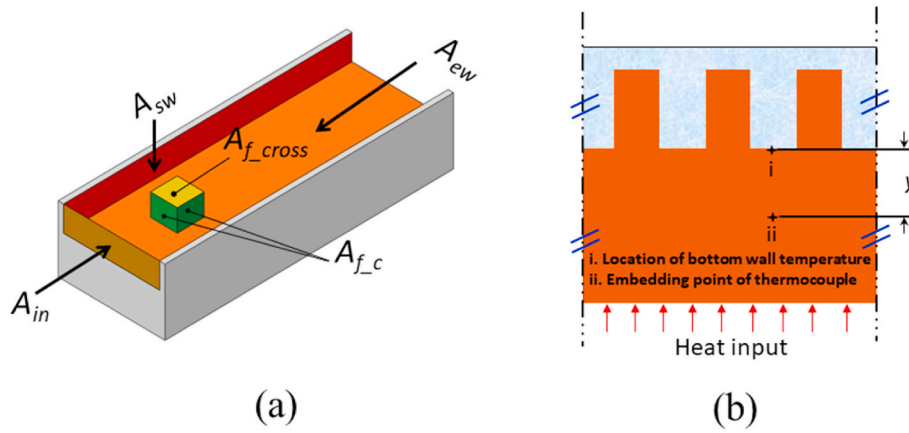


Fig. 6. Representation of (a) different surface areas in the heat sink, (b) temperature measurement technique.

$$Q_{eff} = Q_{total} - Q_{loss} \quad (3)$$

The major cause of heat loss is heat transfer to the surrounding environment. Additionally, available interface between bottom surface of the heat sink and top surface of the heating block hampers the smooth conduction of the heat. Heat losses from the test section has been calculated using the energy balance i.e. difference between total heat generated to the heat carried out by the coolant and is expressed in Eqn. (4). Several authors [42–44] have estimated the heat losses in terms of percentage of the total input power. In the present study also, the average heat loss (Q_{loss}) was obtained from Eqn. (4), after subtracting it from total heat input, effective heat (Q_{eff}) is obtained.

$$Q_{loss} = Q_{total} - \dot{m}C_p(T_{out} - T_{in}) \quad (4)$$

Where, $\dot{m}$ is the fluid flow rate, T_{in} and T_{out} are the temperatures of working fluid at inlet and outlet. After doing the several runs, it was observed that 83–92% of the heat was being carried by the coolant. Hence, for all the heat transfer calculations an average heat loss of 11% of the total input heat is considered. The effective heat flux (q') is expressed on the basis of Q_{eff} and footprint area of the heat sink and is given by:

$$q' = \frac{Q_{eff}}{A_{bs}} \quad (5)$$

The average fluid velocity at the inlet was calculated on the basis of given flow rate ($\dot{m}$) by

$$u_{avg} = \frac{\dot{m}}{\rho_w \times A_{in}} \quad (6)$$

Where ρ_w is density of the deionized water and A_{in} is the inlet cross-sectional area of the heat sinks which is identical for all four configurations. It is worth mentioning that net convective surface area for all four configurations is different. Moreover, due to different geometry and fin tip conditions, calculation of average heat transfer coefficient ($\bar{h}$) needs to be done individually for plain channel, heat sink without TC, and heat sink with available TC.

1. For plain channel, following expression has been used to calculate $\bar{h}$

$$q' A_{bs} = \bar{h} [A_{con-pc}] (\Delta T_{mean}) \quad (7)$$

where A_{con-pc} is the net convective surface area of the plain heat sink which includes the sum of bottom wall and two sidewalls.

$$A_{con-pc} = A_{ew} + 2A_{sw} \quad (8)$$

2. For closed pin fin heat sink ($TC = 0$), fin tip holds adiabatic condition. Hence, the expression may be written as follows.

$$q' A_{bs} = \bar{h} [N_f \eta_f A_{f-c} + (A_{con-pc} - N_f A_{f-cross})] (\Delta T_{mean}) \quad (9)$$

where the fin efficiency for adiabatic conditions is:

$$\eta_f = \frac{\tanh(mH_f)}{mH_f} \quad (10)$$

$$m = \sqrt{\frac{\bar{h} p_{f-cross}}{k_f A_{f-cross}}} = 2\sqrt{\frac{\bar{h}}{k_f W_f}} \quad (11)$$

A_{f-c} is the convective surface area for a single pin fin, which includes four side faces of the pin fin as shown by green colour in Fig. 6(a).

$$A_{f-c} = 4(W_f \times H_f) \quad (12)$$

3. For heat sinks with TC, heat also dissipates from fin tip hence; convective boundary condition at the tip is considered. Accordingly, expression may be written as follows.

$$q' A_{bs} = \bar{h} [N_f \eta_f A_{f-o} + (A_{con-pc} - N_f A_{f-cross})] (\Delta T_{mean}) \quad (13)$$

where A_{f-o} is the wetted surface area of a single fin which includes four sides and top surface.

$$A_{f-o} = 4(W_f \times H_f) + A_{f-cross} \quad (14)$$

$A_{f-cross}$ is pin fin cross sectional area as depicted by yellow colour in Fig. 6(a). The efficiency is calculated considering convective fin top boundary condition and expressed as follows.

$$\eta_f = \frac{\sqrt{\bar{h} p_{f-cross} k_f A_{f-cross}} \left[\frac{\tanh(mH_f) + \frac{\bar{h}}{k_f m}}{1 + \frac{\bar{h}}{k_f m} \tanh(mH_f)} \right]}{\bar{h} (p_{f-cross} H_f + A_{f-cross})} \quad (15)$$

In all the above expressions, ΔT_{mean} is the average temperature difference between the heat sink bottom wall and the working fluid which is obtained as per Eqn. (16).

$$\Delta T_{mean} = \bar{T}_{bw} - \bar{T}_{bf} = \frac{1}{2}(T_{bw1} + T_{bw2}) - \frac{1}{2}(T_{in} + T_{out}) \quad (16)$$

Where, subscript 1 and 2 are the bottom wall temperatures at two different locations (as mentioned above 1st and 2nd Tc is embedded at a distance of 5 and 22 mm respectively from inlet of the heat sink) in the heat sink, $\bar{T}_{bw}$ and $\bar{T}_{bf}$ are average bottom wall temperature and average

Table 4
Uncertainty associated with important parameters.

Parameters	Percentage uncertainty
Physical dimension	±1.33
Q	±1.59
Q	±2.46
q'	±3.00
$\bar{h}$	±3.16
$\overline{Nu}$	±3.68
Pressure drop (ΔP)	±2.12
TPF	±5.3

bulk fluid temperature.

It should be noted that bottom wall temperature (T_{bw}) of the microchannel is calculated considering the one-dimensional heat conduction amid channel's bottom wall i.e. point i and embedding point of thermocouples (point ii) as shown in Fig. 6(b). Thermocouple is inserted 2 mm ($y = 2$ mm) below to the point i . Hence, T_{bw} has been calculated as follows

$$T_{bw} = T_{ii} - \frac{(q' y)}{k_{cu}} \quad (17)$$

In this equation, T_{ii} is the temperature at embedding point and k_{cu} is copper thermal conductivity.

By knowing heat flux, $\bar{T}_{bw}$ and $\bar{T}_{bf}$, iterative technique has been used for evaluating heat transfer coefficients using a MATLAB program. Calculation of Re has been made on the basis of channel inlet hydraulic diameter.

$$Re = \frac{\rho_w \times u_{avg} \times D_h}{\mu_w} \quad (18)$$

Where μ_w is the dynamic viscosity of the fluid and D_h is the channel inlet hydraulic diameter which was calculated from Eqn. (19)

$$D_h = \frac{2W_{ch}H_{ch}}{W_{ch} + H_{ch}} \quad (19)$$

where, W_{ch} and H_{ch} are channel width and height as represented in Figs. 2 and 3. Further, average Nusselt number ($\overline{Nu}$) has been calculated from the following equations.

$$\overline{Nu} = \frac{\bar{h}D_h}{k} \quad (20)$$

where k is the thermal conductivity of the coolant water.

Thermal performance factor (TPF) has been [9,26,45] used to estimate the overall thermal performance of the modified heat sinks. As mentioned below Eq. (21), it includes the heat transfer capacity as well as pressure drop that encounter in the modified heat sink with respect to the base heat sink. In the present case, plain channel has been taken as the base heat sink.

$$TPF = \frac{\overline{Nu}/\overline{Nu}_{plain}}{(\Delta p/\Delta p_{plain})^{1/3}} \quad (21)$$

In the above equation, subscript *plain* represents the case of plain microchannel heat sink having no fins.

2.5. Uncertainty analysis and repeatability

It is understood that experimental values have certain errors associated with it, uncertainty analysis provides the maximum deviation in the measurements [46]. Before execution of the experiments, all the measuring equipment's were carefully calibrated. Peristaltic pump shows the uncertainty of ±1.56% in volume flow rate. Uncertainties in the temperature (using K-type thermocouple) and pressure

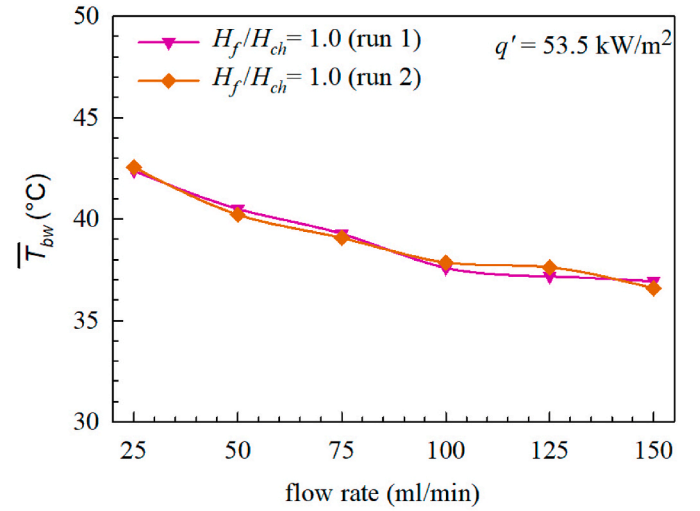


Fig. 7. Repeatability of $\bar{T}_{bw}$

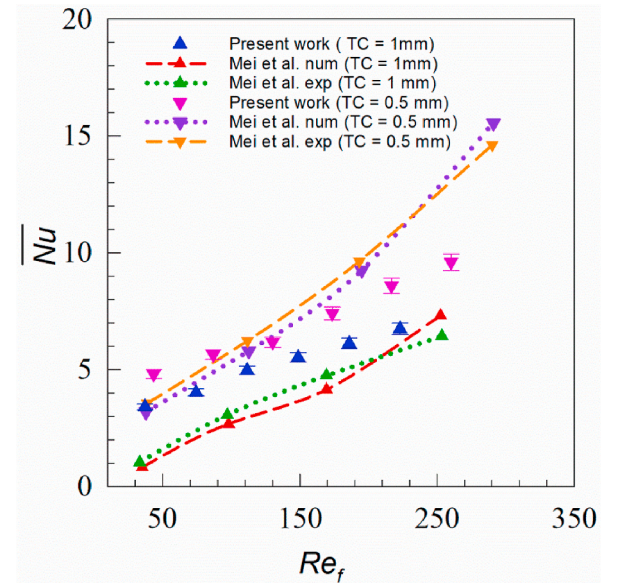


Fig. 8. Comparison of $\overline{Nu}$ with Mei et al. [39].

measurement are ±1.0% and ±1.5% respectively. Similarly, maximum error occurred in the measurement of physical dimensions, voltage and current is ±1.33%, ±0.69 and ±1.43% respectively. Uncertainties in all the derived parameters were calculated by the expressions given in the literature [47,48]. Derived parameters with their uncertainties are tabulated in Table 4. It should be noted that maximum uncertainty of 5.3% occurs in the calculation thermal performance factors (TPF). Repeatability of the experiments has also been checked and one of observations (at $q' = 53.5$ kW/m²) of $\bar{T}_{bw}$ was observed during both runs conducted at different times.

2.6. Validation of the work

A comparison of $\overline{Nu}$ obtained from the present work with an uncertainty of ±3.68% and that of Mei et al. [36] has been shown in Fig. 8. It is worth mentioning that Mei et al. [39] have studied the effect of TC of 0.5 mm and 1.0 mm for circular pin fin heat sink with total number of 500 fins. They considered the D_f of 1 mm and channel width of 0.8 mm.

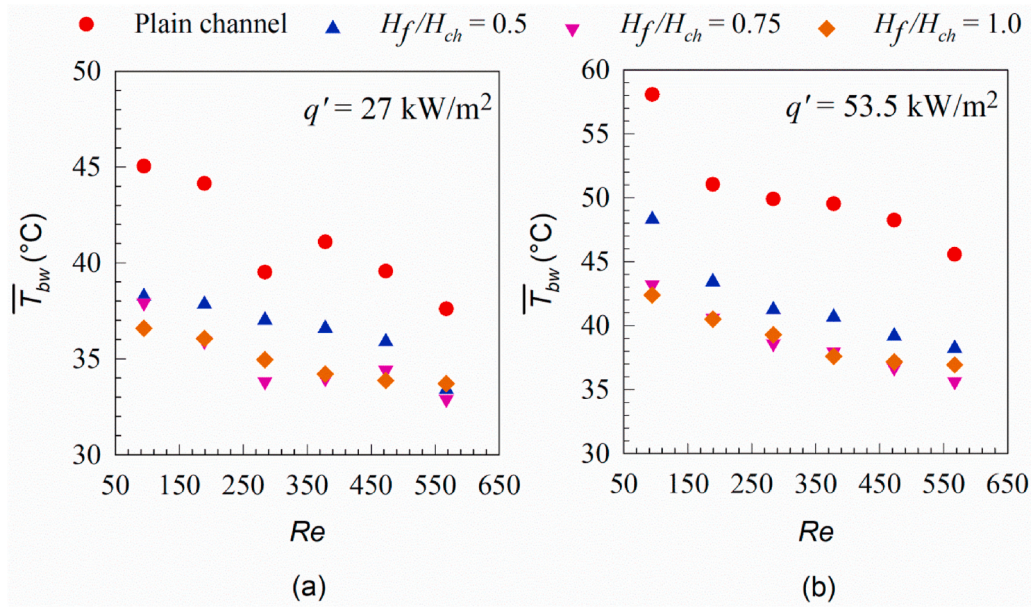


Fig. 9. Variation of $\bar{T}_{bw}$ with Re at (a) $q' = 27 \text{ kW/m}^2$ (b) $q' = 53.5 \text{ kW/m}^2$.

Fins were arranged in an inline manner. Maximum channel height was similar to present study i.e., 2 mm.

In order to compare the Nu , different sets of experiments were conducted in the comparable operating ranges. Besides, the calculation of Re is done in the similar manner (done by Mei et al. [39]) which is based on D_f and represented as Re_f . Comparison of the present result clearly indicates that increasing trend of $\bar{Nu}$ with Re_f is nearly alike as that of Mei et al. [39]. Evidently, with reducing TC from 1.0 to 0.5 mm, $\bar{Nu}$ increases which is in accordance to the findings of the reference paper. However, with varying Re_f an average deviation of $\approx 15\text{--}20\%$ appears in $\bar{Nu}$. It is due to the prevailing difference in geometries (of present heat sink and Mei et al. [39]) mainly in terms of number of fins, its shape and overall size of the heat sink. Also, they vary the channel height keeping the fin height constant, while in the present study, fin height was varied from 1 mm to 2 mm.

3. Results and discussion

Results have been illustrated in a comparative manner for all four considered heat sink configurations. Deionized water at constant inlet temperature of 30°C was used during all the experiments. Important findings such as $\bar{T}_{bw}$, $\bar{h}$, $\bar{Nu}$, total pressure drop (ΔP) and TPF have been thoroughly discussed. Coolant flow characteristics have been explained through the velocity contour obtained from numerical simulations. The present study was performed for six different values of coolant mass flow rate ranging from 25 to 150 ml/min (corresponding mass flux is $G = 23\text{--}138 \text{ kg/m}^2$) in the step of 25 ml/min. Laminar flow prevails in the heat sink where Re varies from 90 to 570 and range of applied heat flux is $q = 30\text{--}60 \text{ kW/m}^2$.

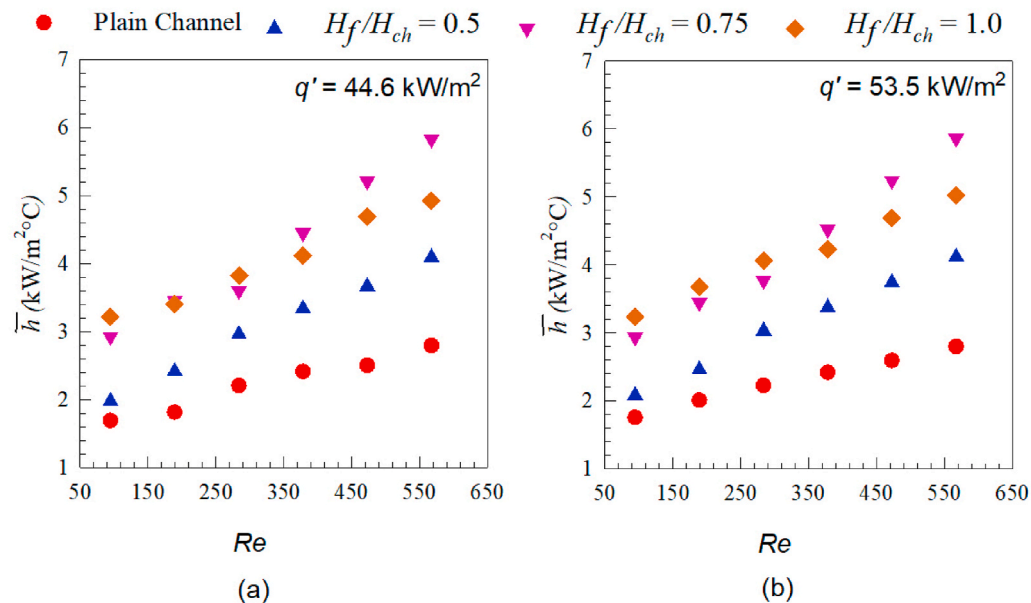


Fig. 10. Variation of $\bar{h}$ with Re at (a) $q' = 44.6 \text{ kW/m}^2$ (d) $q' = 53.5 \text{ kW/m}^2$.

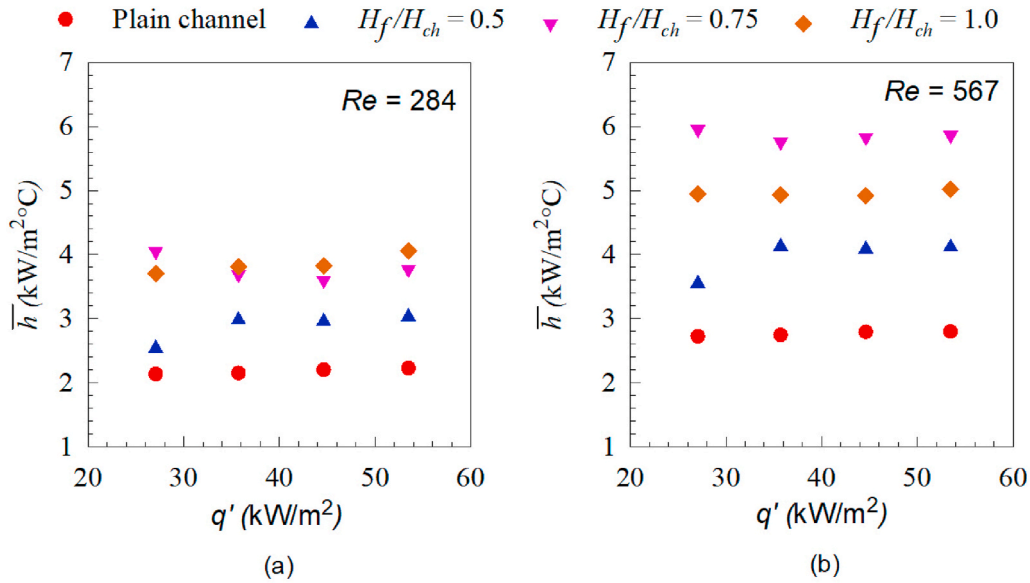


Fig. 11. Effect of q' on $\bar{h}$ at (a) $Re = 284$ (b) $Re = 567$.

3.1. Heat transfer characteristics

3.1.1. Bottom wall temperature

$\bar{T}_{bw}$ for all the four configurations have been shown in Fig. 9 (a & b). Variation of $\bar{T}_{bw}$ with respect to Re at $q' = 27 \text{ kW/m}^2$ has been illustrated in Fig. 9(a). Irrespective of channel configuration, $\bar{T}_{bw}$ decreases almost linearly with Re . It is obvious because with increasing Re , coolant mass flow rate increases which carry away more heat from the heat sink. Figure further reveals that $\bar{T}_{bw}$ of the plain channel is significantly high compared to finned configurations. Slope of the curve is also steeper with maximum and minimum temperature values of 45 °C and 38 °C respectively at $Re = 94$ and 567 . Among the finned configurations, it can be clearly observed that heat sink with fin height of 1 mm consistently holds higher, $\bar{T}_{bw}$ nearly at all Re . Whereas, $\bar{T}_{bw}$ for the remaining two configurations is very close to each other. However, it may be observed that heat sink of $H_f = 1.5 \text{ mm}$, has slightly lower $\bar{T}_{bw}$ explicitly at higher values of Re . As shown in Fig. 9 (b), at higher value of $q' = 53.5 \text{ kW/m}^2$, characteristics of the curves are almost similar as that of Fig. 9(a). $\bar{T}_{bw}$ corresponding to $H_f/H_{ch} = 0.75$ and $H_f/H_{ch} = 1.0$ is in close proximity and consistently lower compared to remaining two profiles. It is understood that lower values of $\bar{T}_{bw}$ means higher heat dissipation from the particular heat sink. Evidently, plain channel is found with lowest heat transfer capacity followed by $H_f/H_{ch} = 0.5$. Availability of TC has shown its importance which is clearly visible from the values of $\bar{T}_{bw}$. Higher value of TC (as in the case of $H_f/H_{ch} = 0.5$) is not worthy in respect of the heat dissipation potential primary due to considerably low convective surface area. Although if a small TC is provided as deliberated from $H_f/H_{ch} = 0.75$, one can achieve lower substrate temperature due to higher heat dissipation. A similar type of observation was also reported by Far et al. [41] although they have worked with coolant as nano encapsulated phase change material slurry. As it may be seen in Fig. 9(b), minimum attainable value of $\bar{T}_{bw}$ is $\approx 35.5 \text{ °C}$ at $Re = 567$. Furthermore, both Fig. 9 (a & b) broadly reveal that $\bar{T}_{bw}$ is generally lower for $H_f/H_{ch} = 0.75$. Although existing differences are very marginal between $H_f/H_{ch} = 0.75$ and $H_f/H_{ch} = 1.0$.

3.1.2. Heat transfer coefficient

In order to comprehend the heat transfer rate from the heat sinks, $\bar{h}$ has been drawn with varying Re . Fig. 10 (a & b) shows $\bar{h}$ corresponding to $q' = 44.6 \text{ kW/m}^2$ and 53.5 kW/m^2 respectively. It should be noted

that determination of $\bar{h}$ has been done as per Eqns. (7), (9) and (13). Irrespective of q , almost linear increasing trend of $\bar{h}$ has been found with Re . Also, $\bar{h}$ is negligibly affected with changing values of heat fluxes. As shown in Fig. 10 (a), plain channel exhibits minimum $\bar{h}$ for all the values of Re . Moreover, slope of the curve is less steep compared to other configurations and maximum reported value of $\bar{h}$ is 2.8 kW/m^2 at $Re = 567$. Undoubtedly, pin fin heat sinks support higher heat transfer coefficients in comparison to plain heat sink, it is attributed to higher convective surface area and different coolant flow behavior evolved due to the presence of pin fins with TC. Existence of fins facilitates enhanced fluid mixing by providing secondary or interconnecting flow passage along the transverse directions. Among the finned heat sinks, $H_f/H_{ch} = 0.5$ imitates consistently lower values of $\bar{h}$ but, compared to plain microchannel, $\bar{h}$ is $\approx 32\%$ higher corresponding to maximum value of Re and $q' = 44.6 \text{ kW/m}^2$.

In all the cases, $\bar{h}$ increases with Re , possibly it is due to the two prominent reasons. (i) Increasing Re causes higher coolant flow rate resulted in more heat carried out by the fluid. (ii) Thermal developing entrance length elongates with Re and certainly, longer developing zone supports more heat transfer rate. Since overall length of the microchannel heat sink is considerably small hence, even slight change in developing entrance length is significant in view of heat transfer rate. Moreover, in finned channels, higher Re also induces more lateral direction coolant flow and interactions causing improved fluid mixing which supports better heat transfer.

With increasing H_f/H_{ch} to 0.75 , heat transfer rate significantly increases and attains maximum $\bar{h}$ specifically at higher Re . Although at smaller values of Re , heat sink with $H_f/H_{ch} = 1.0$ yields higher heat transfer coefficients compared to $H_f/H_{ch} = 0.75$. It has been found that beyond $Re \approx 350$, heat transfer coefficient in $H_f/H_{ch} = 0.75$ shoots up and becomes maximum compared to all the configurations. As shown in Fig. 10(b), maximum attained value of $\bar{h}$ is 5.8 kW/m^2 for $H_f/H_{ch} = 0.75$ at $Re = 567$. It is important to point out that both configurations of the heat sinks having $H_f/H_{ch} = 0.75$ and 1.0 hold nearly equal convective surface areas nevertheless, heat transfer performance of the heat sink with $H_f/H_{ch} = 0.75$ is comparatively better. It is because of available TC of 0.5 mm which provides distinct coolant flow behavior beyond $Re = 350$. Unlike $H_f/H_{ch} = 1.0$, in this case, top surface of the fins is also exposed to the coolant and participating in the heat exchange process.

The effect of heat flux on $\bar{h}$ is shown in Fig. 11 (a & b). It is apparent

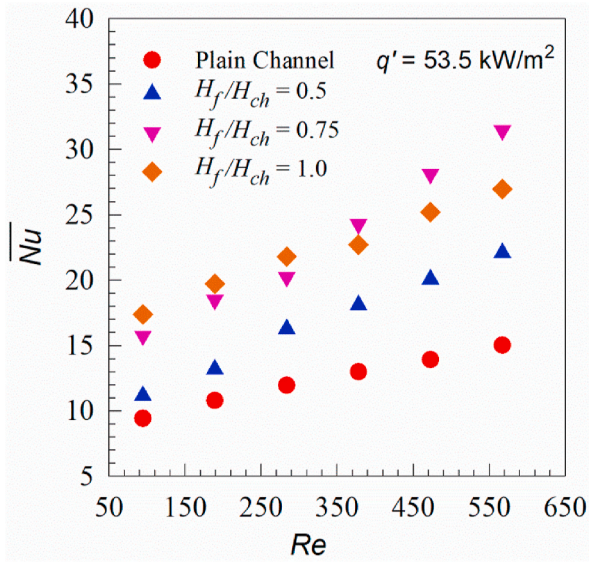


Fig. 12. Variation of $\overline{Nu}$ with Re for all the configurations at $q' = 53.5 \text{ kW/m}^2$.

from the figures that for a constant value of Re , $\bar{h}$ remains almost same at all applied heat fluxes. Hence, it may be concluded that $\bar{h}$ is negligibly affected by the heat flux. Close observation of both the figures also reveals that at $Re = 284$, pin fin configurations of $H_f/H_{ch} = 0.75$ and 1.0 are showing nearly similar $\bar{h}$ for all the heat flux ranges. However, at $Re = 567$, heat sink with $H_f/H_{ch} = 0.75$ significantly dominates in terms of improved heat transfer rate and it has attained $\approx 15\text{--}16\%$ higher $\bar{h}$ compared to $H_f/H_{ch} = 1.0$.

Fig. 12 shows the variation of $\overline{Nu}$ with Re for the effective heat flux of 53.5 kW/m^2 . $\overline{Nu}$ has been calculated in accordance to Eqn. (20). It can be seen that the characteristics of the $\overline{Nu}$ curves are consistent with $\bar{h}$ as discussed in the previous section. Plain channel shows the lowest values of $\overline{Nu}$ ranging from 9 to 15 respectively at Re of 94 and 567 and $q' = 53.5 \text{ kW/m}^2$. Compared to plain channel, heat sink with $H_f/H_{ch} = 0.5$ attains higher $\overline{Nu}$ ranging from 11 to 22. Additionally, obtained data points form a steeper slope compared to plain channel which is more consistent to the slopes of finned configurations. It indicates that finned profiles are having better heat transfer rate with increasing Re . $\overline{Nu}$ further increases with increasing the H_f to 1.5 mm and subsequently to 2.0 mm .

Both of these configurations of the heat sink show $\overline{Nu}$ very close to each other. However, at higher Re beyond 350, pattern of $\overline{Nu}$ values alters and $H_f/H_{ch} = 0.75$ attains the maximum values of $\overline{Nu}$ amongst all the considered heat sink configurations. This profile of the channel delivers maximum value of $\overline{Nu}$ i.e., 31 at $Re = 567$ which is even $\approx 12\%$ higher than the heat sink of $H_f/H_{ch} = 1.0$. Prajapati [32], Mei et al. [39] and Rozati et al. [40] have also reported that optimum TC ($\approx 10\text{--}20\%$) supports better heat transfer rate compared to completely closed heat sink.

It is important to highlight that convective surface area (solid-liquid contact surface) of the heat sinks with $H_f/H_{ch} = 0.75$ and 1.0 is 639 mm^2 and 687 mm^2 respectively. Interestingly, even after having higher convective surface area in the case of $H_f/H_{ch} = 1.0$, its heat transfer performance is not much encouraging as compared to $H_f/H_{ch} = 0.75$. This fact confirms that convective surface area is not a sole factor responsible to enhance the heat transfer rate. Rather, evolution of favorable coolant flow behavior is equally important and vital to support better thermal performance. In the present investigation, it has been observed that in the heat sink without any TC ($H_f/H_{ch} = 1.0$) coolant is restricted to flow through the channel passages only however, in the heat sink with TC = 0.5 mm ($H_f/H_{ch} = 0.75$) certain amount of coolant also sips through the available opening at the top. Consequently, in addition to channel flow passages (through which bulk fluid flows), small fraction of coolant also receives the exposure of fin top surface and open space. Two distinct orientations/streams of fluid flow especially at higher Re , evolve immature or undeveloped boundary layer, which increases three dimensionalities in flow, as a result fluid mixing is augmented that facilitates improved heat transfer rate in the configuration having $H_f/H_{ch} = 0.75$. Besides, coolant flow behavior is thoroughly discussed in the upcoming section with the help of velocity contour predicted from the numerical work.

3.2. Pressure drop

The pressure difference between inlet and outlet estimates the pumping power needed to maintain the flow of working fluid through the heat sink. Fig. 13 shows the variation of pressure drop for different configurations of the heat sink at effective heat flux of 53.5 kW/m^2 . It is worth mentioning that two independent pressure transducers are used to measure the pressure in the inlet and outlet plenums separately. Pressure transducers generate signals in mA which is calibrated to corresponding pressure data. Total pressure drop (ΔP) is the difference of measured pressure values in inlet and outlet plenums. It also includes the minor pressure losses occurring in swallow passage, sudden

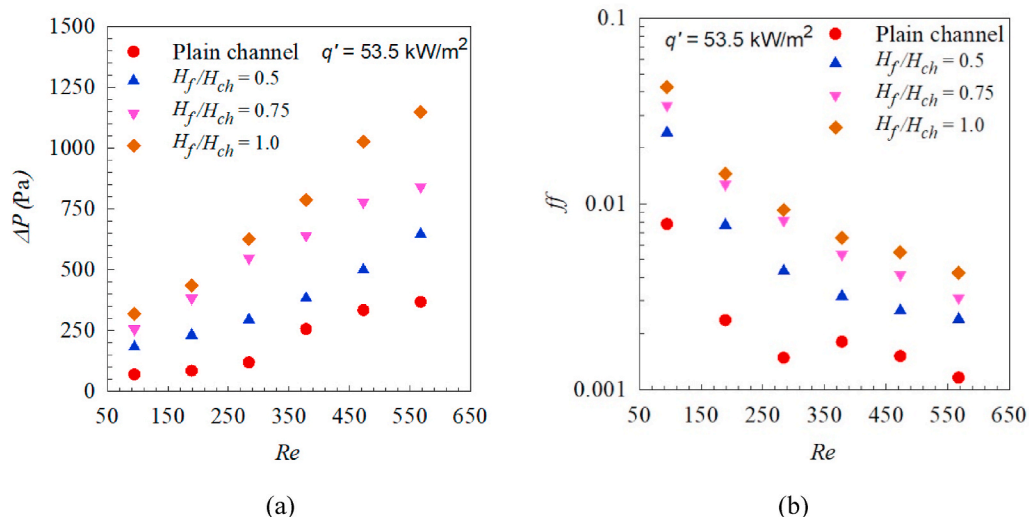


Fig. 13. Variation of (a) pressure drop and (b) friction factor with Re at $q' = 53.5 \text{ kW/m}^2$.

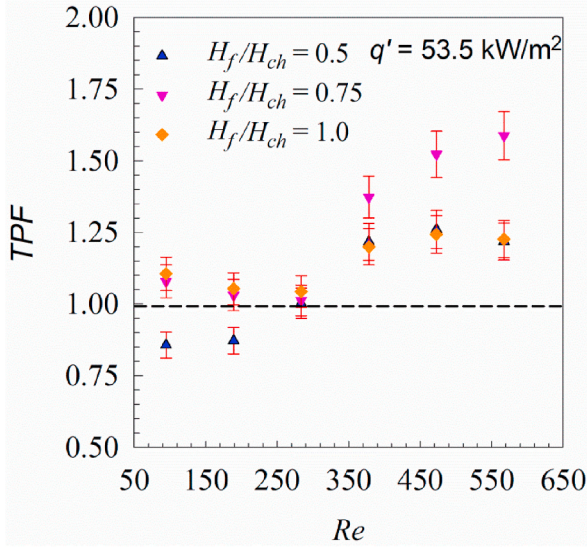


Fig. 14. Variation of TPF with Re for all the configurations at $q' = 53.5$ kW/m².

contraction and enlargement at the inlet and outlet sections of the channel. The value of total minor losses varies between 5 and 25 Pa for all the flow rates of the coolant. Compared to total pressure drop in the test section, this value is considerably less hence, it has been neglected in the calculations [42,43,49].

Irrespective of the channel configurations, it is obvious that pressure drop significantly increases with Re . Since single phase liquid flows inside the channel, almost linear pressure drop may be observed in all the heat sinks. It is apparent that among the entire four configurations, plain channel provides minimum pressure drop as there is smooth flow passage for the coolant. Range of pressure drop measured in the plain channel is merely ≈ 55 – 362 Pa for all Re values. Presence of pin fins in the heat sink creates flow obstruction and resistance to fluid flow hence, higher pressure drop encounters in the finned heat sinks. As shown in Fig. 13, longer the fin height more is the flow resistance accordingly;

pressure drop also increases in the channel. Subsequently, heat sink of pin fin $H_f/H_{ch} = 1.0$ offers maximum pressure drop followed by $H_f/H_{ch} = 0.75$ and $H_f/H_{ch} = 0.5$. At the effective heat flux of 53.5 kW/m², maximum measured pressure drops are ≈ 1150 Pa, 840 Pa and 650 Pa respectively at $Re = 567$.

Influence of the heat flux on pressure drop is found to be negligible in the present operating ranges. Furthermore, it is clear that compared to closed channel, open microchannels with TC has smaller pressure drop. It may be attributed to following two reasons (i) presence of smaller fin height causing less resistance to flow, (ii) open space offers less constriction and flow is mostly undisturbed. In addition to pressure drop, friction factor has also been estimated [50] as shown in Fig. 13(b). It may be observed that at $q' = 53.5$ kW/m², frictional resistance decreases in all the heat sinks with increasing value of Re . Slope of the curves reduces significantly at higher values of Re . Further, it is obvious that the plain channel configuration exhibits the lowest friction factor because no obstructions are present in the flow field. Among the finned configurations, $H_f/H_{ch} = 0.5$ and $H_f/H_{ch} = 1.0$ have shown minimum and maximum ff respectively. Owing to longer fins, heat sink with $H_f/H_{ch} = 1.0$ offers more flow obstruction in the flow path, leading towards higher friction resistance. Nevertheless, $H_f/H_{ch} = 0.75$ deliberates ff slightly lesser than $H_f/H_{ch} = 1.0$.

3.3. Thermo-hydraulic performance

It has been commonly observed that whenever an attempt is made to enhance the heat transfer rate in microchannel heat sink, usually pressure drop also increases. Estimation of the TPF , as expressed in Eq. (21), quantifies the relative increment in overall thermal performance of the modified channels in comparison to the base channel. Fig. 14 illustrates the calculated TPF for all the heat sinks at $q' = 53.5$ kW/m² with varying Re . It is desirable that TPF should be greater than 1 to justify if any enhancement in thermal performance is made in the proposed configurations. Accordingly, it may be appreciated in the figure that except for $H_f/H_{ch} = 0.5$, both configurations of $H_f/H_{ch} = 0.75$ and 1.0 show TPF greater than 1. Even $H_f/H_{ch} = 0.5$ is having $TPF > 1$ except at two values of Re i.e., 94 and 189.

It may be observed that at the lower Re , TPF of $H_f/H_{ch} = 0.75$ and 1.0

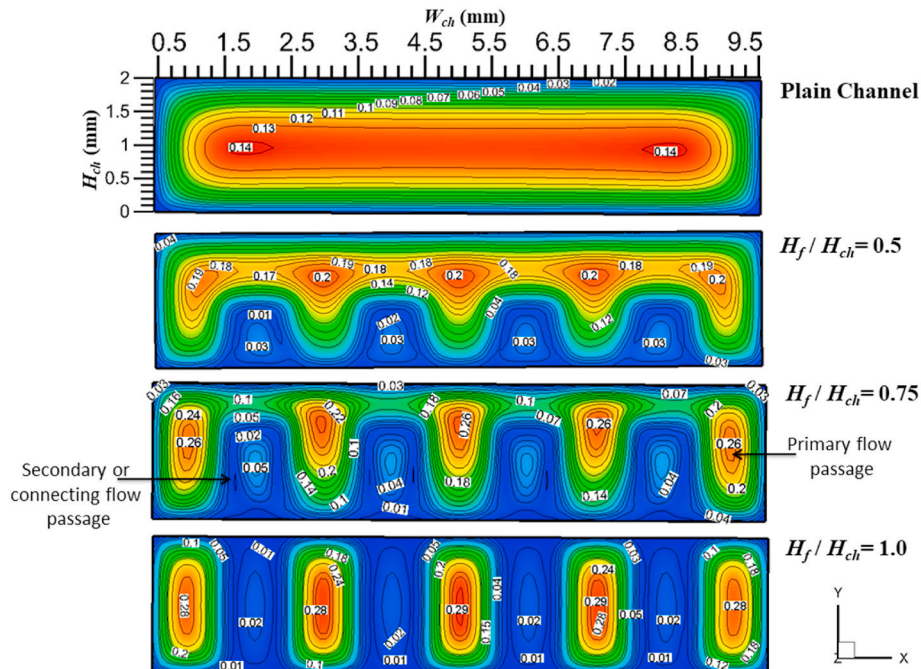


Fig. 15. Velocity contour at the mid plane along the channel length ($z = 13.5$ mm from inlet) at $Re = 378$ and $q' = 53.5$ kW/m².

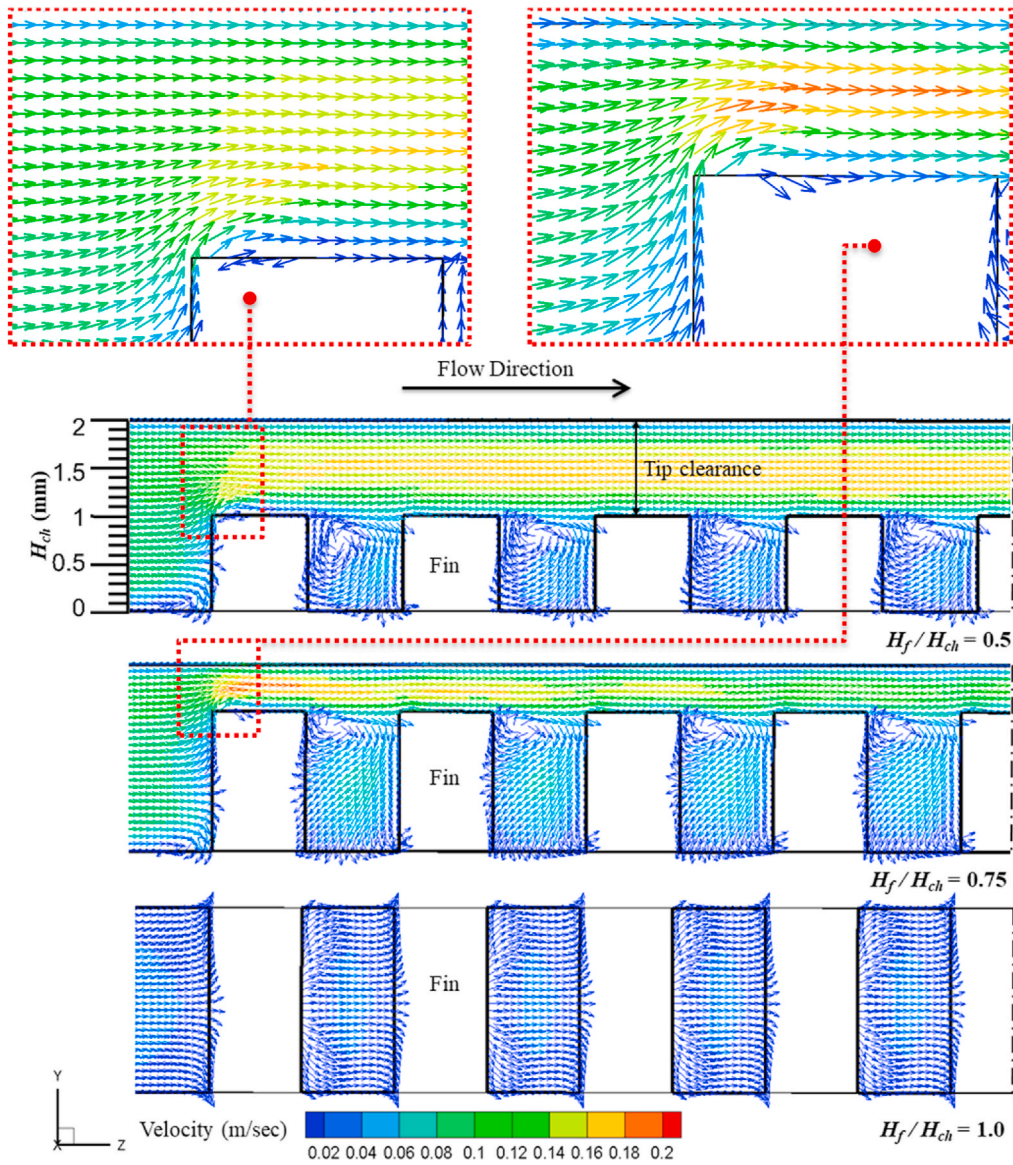


Fig. 16. Velocity vector at a plane ($x = 6$ mm from left side wall) at $Re = 378$ and $q' = 53.5$ kW/m².

is comparable however, beyond Re of 350, $H_f/H_{ch} = 0.75$ deliberates considerably higher TPF which approaches more than 1.5. While, the configurations $H_f/H_{ch} = 0.5$ and $H_f/H_{ch} = 1.0$ have yielded almost similar performance for Re greater than 350. A jump in TPF values (for $H_f/H_{ch} = 0.75$) beyond $Re = 350$ is consistent with the heat transfer augmentation pattern as discussed previously. This may be attributed to the higher growth in heat transfer rate as compared to the pressure drop due to favorable flow behavior. Besides, at the higher Re , more heat is dissipated through the convection and coolant flow becomes scattered in the region of tip clearance. Additionally, secondary flow passages of the heat sink allow better mixing of the coolant. Due to these favorable circumstances, pin fin heat sink with fin height equals to 0.75 times of channel height has superior overall thermal performance compared to its counterparts considered in this study.

3.4. Coolant flow characteristics

In order to comprehensively understand the fluid flow behavior in all the four configurations of the heat sink, three-dimensional numerical simulation has also been carried out using ANSYS Fluent V18.0. Some numerical findings, explicitly, velocity contours are shown here to

visualize the results with better insight and explanation. However, for complete details of the numerical investigation, our previous work [9] may be referred. In this section, a simplified computational domain without considering the base thickness (H_b) of the heat sinks is numerically simulated nevertheless, remaining dimensions and operating conditions ($Re = 378$ and $q' = 53.5$ kW/m²) are identical to the actual heat sink used in the experimental work. Computational domains are meshed with structured hexahedral volume elements. The mesh size has been optimized by grid independence test and grid size of 0.075 mm was used for all the simulations. Velocity inlet and pressure outlet boundary conditions are imposed at the inlet and out of the heat sinks. In accordance to the experiments, laminar flow is considered in the simulation and SIMPLE algorithm has been used to solve the pressure-velocity coupling. To solve the conduction and convection heat transfer simultaneously in solid and fluid zones, conjugate method has been used and thermophysical properties of the coolant i.e., liquid water is varied with temperature. Each case approximately takes ≈ 30 min of physical time to get steady state converged solution.

Fig. 15–17 show the coolant velocity profile at different planes of the heat sinks. Fig. 15 illustrates the velocity contours at the mid-fluid plane along the channel length at a distance of $z = 13.5$ mm from the inlet. The

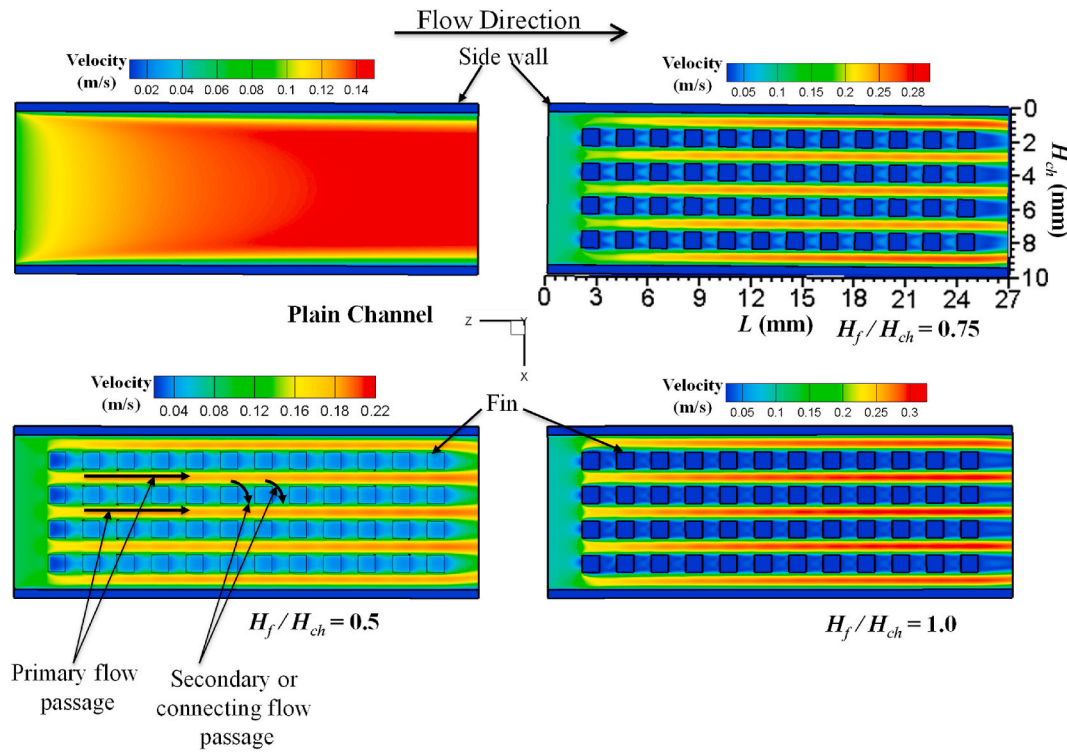


Fig. 17. Velocity contour at the mid fluid plane ($y = 1$ mm from top) corresponding to $Re = 378$ and $q' = 53.5$ kW/m².

considered plane is located in between two successive fin rows. The velocity contour is drawn corresponding to $Re = 378$ and $q' = 53.5$ kW/m². All four heat sinks are clearly configured from the colour contour of the velocity profile. Plain channel has no fin hence, there is not any flow obstruction in the flow field consequently, symmetric velocity distribution evolves in this channel with maximum velocity at the middle of the heat sink. Availability of the fins in rest of the configurations significantly alters the velocity distribution and propagates favorable flow behavior that supports better heat transfer rate. In Fig. 15 it can be seen that due to presence of larger TC for $H_f/H_{ch} = 0.5$, mainstream of the coolant flow is prominently oriented in the available open space at the top. Fluid flow also prevails in primary flow passages however, secondary flow passages observe vortex formation and allow fluid mixing up to certain extent only. With increasing fin height to $H_f = 1.5$ mm ($H_f/H_{ch} = 0.75$), fluid flow behavior slightly changes where major fraction of bulk fluid flows through the channel passages. Moreover, it can be seen that due to confined open space, flow is more distributed and dispersed causing enhanced and efficient solid-fluid interaction. Additionally, secondary flow passage is comparatively more participative and profound eddies formation occurs in this profile. Hence, fluid mixing is maximum in this configuration which leads to superior performance of the heat sink with $H_f/H_{ch} = 0.75$. It is interesting to note down that net convective surface area for $H_f/H_{ch} = 0.75$ is 639 mm² compared to 687 mm² for $H_f/H_{ch} = 1.0$, even then due to favorable flow behavior, heat sink with $H_f/H_{ch} = 0.75$ deliberates better heat transfer rate. Fluid flow in the closed heat sink ($H_f/H_{ch} = 1.0$) is confined to channel passages only therefore, coolant get less exposure to disperse in the lateral directions and its propagation is limited. Consequently, even after possessing higher convective surface area, its heat transfer performance is not encouraging and slightly lesser than heat sink of $H_f/H_{ch} = 0.75$. It may be concluded that open space above the fin's top surface provides a connecting link between primary and secondary flow passages and optimum value of TC is desirable to propagate enhanced fluid mixing heat transfer rate.

In order to further visualize the flow pattern, velocity vector plots for finned configurations are shown in Fig. 16. The plots are drawn at a

plane parallel to Y-Z axis and located at $x = 6$ mm from left side wall. Considered plane traverse through the fins present in third column at the mid-section. In continuation to earlier discussion, it can be observed that when the H_f/H_{ch} is increased from 0.5 to 0.75, confinement effect in the open space is more that resulted in comparatively higher flow velocity in the channel which if further confirmed from the enlarged view of the vector plots. Moreover, due to obstacle created by the fins in the flow path, recirculating vortices are formed at the trailing edge (in the secondary flow passage) of each fin. Characteristics of the vortices are almost similar in both the configuration with TC however, different in closed heat sink ($H_f/H_{ch} = 1.0$). Close observation of the vector plot for $H_f/H_{ch} = 1.0$ reveals that coolant trapped between two successive fins is unable to create any circulation and seldomly contribute in coolant mixing process hence, heat transfer rate is not much encouraging in this configuration.

Fig. 17 captures the flow behavior of the coolant in a plane located at $y = 1$ mm from the top surface. Relative position of fins, side wall and flow passages can be clearly visualized from Fig. 17. Plain channel has no fins hence; velocity contour is entirely different whereas, remaining heat sinks show nearly identical pattern of velocity distribution. It can be seen that due to inline arrangement of pin fins, fluid prominently flows through the primary passages and recirculating vortices are originated in the connecting passages. Among all the configurations, heat sink with having $H_f/H_{ch} = 0.75$ experiences high velocity vortices thereby, allowing increased lateral coolant flow and better fluid mixing. Due to prevailing favorable flow behavior, this configuration has shown superior heat transfer rate compare to its counterpart.

4. Conclusions

Present study examines the overall thermal performance of four different configurations of the copper heat sink. Three configurations incorporate square pin fins with varying TC whereas an additional plain channel has also been considered for comparison of results. Experiments were conducted using single phase liquid water as working medium at a fixed inlet temperature of 30 °C. Applied ranges of the heat flux and Re

are 30–60 kW/m² and 90–570 respectively. Influence of TC is investigated and results have been illustrated through comparative analysis of $\bar{h}$, $\overline{Nu}$, ΔP and TPF . Numerical study has also been carried out to understand the fluid flow characteristics in the heat sinks with the help of velocity contours. Following essential conclusions may be drawn from the present study.

1. Availability of TC is sensitive to heat transfer rate as well as pressure drop and it certainly influences the thermal performance of the microchannel heat sink. Presence of TC provides additional space for the coolant to flow and promotes dispersed flow and augmented fluid mixing.
2. Heat transfer potential with $H_f/H_{ch} = 0.5$ i.e. $TC = 50\%$ is not worthy as it leads to considerably lower convective surface area. Whereas, TC of 25% has shown superior performance.
3. Compared to closed configuration i.e., $H_f/H_{ch} = 1.0$, heat sink with $H_f/H_{ch} = 0.75$ ($TC = 25\%$) has $\approx 7\%$ lesser convective surface area, even then it has shown better heat transfer rate explicitly at the higher values of $Re \geq 350$. Possible reason of better performance is understood through the numerical result which shows that the heat sink with $TC = 25\%$ facilitates dispersed fluid flow behavior in the confined space of the TC which propagates thinner boundary layer, three dimensionalities in flow and augmented fluid mixing.
4. Heat sink with $TC = 25\%$ has deliberated $\overline{Nu} \approx 10\text{--}12\%$ and $TPF \approx 15\%$ higher as compared to heat sink with $TC = 0$ explicitly at $Re >$

350. Hence, selection of optimum value of TC in the range of 20–25% of channel height is vital.

5. The pumping power increases with increasing fin height; consequently, plain channel has minimum pressure drop followed by $TC = 50\%$. Heat sink with $TC = 25\%$ offers 19% smaller pressure drop compared to closed configuration which experiences maximum flow resistance.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Nomenclature

A	area, mm ²
C_p	specific heat, J/kg K
D_f	pin fin diameter, mm
D_h	hydraulic diameter, mm
DC	direct current
ff	friction factor
G	mass flux, Kg/m ² s
$\bar{h}$	average heat transfer coefficient, W/m ² °C
H	height, mm
I	current, Amp
k	thermal conductivity, W/mK
L	length, mm
$\dot{m}$	fluid flow rate, Kg/s
N_f	total number of fins
$\overline{Nu}$	Nusselt number
p	perimeter, mm
P	pressure, Pa
Q	heat, W
q	heat flux, kW/m ²
q'	effective heat flux, kW/m ²
Re	inlet Reynolds number
W	width, mm
$\bar{T}$	average temperature, °C
TC	tip clearance
T_c	thermocouple
TPF	thermal performance factor
u	fluid velocity, m/s
V	voltage, Volt
W	width, mm

Greek symbols

ρ	density, kg/m ³
μ	viscosity, kg/m-s
η	efficiency

Subscript

avg	average
b	heat sink base
bs	bottom surface area or footprint area
bw	bottom wall
bf	bulk fluid
c	individual channel
con _{pc}	convective surface area of the plain heat sink
ch	channel
cu	copper
eff	effective
ew	end wall
f	fin
f _c	fin surface area excluding top and bottom face
f _{cross}	fin cross-section
f _o	fin wetted surface area
loss	loss
in	inlet
out	outlet
sw	side wall
s	solid
w	water

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